

Industry 4.0

Complete student lesson for course code **4339**.

Revision 2026 Semester 4 Program Core 4 modules

Course snapshot

Scheme: 3 (L:0,T:0,P:3); 45 periods

Credits: 0

Contact: 45 periods per semester

Module hours listed: 43

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

4339

Semester

4

Credits

0

Teaching scheme

3 (L:0,T:0,P:3); 45 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Origin, concepts and framework of Industry 4.0	11	CO1
2	Technologies, connectivity and PLC programming	11	CO2
3	Building blocks, CPS, IIoT and RAMI 4.0	11	CO3
4	Robotics in Industry 4.0	10	CO4

Prerequisites

- Basic Engineering Physics, sensors and actuators, computer skills and Python programming.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. Learn and understand the basics of Industry 4.0.
2. Develop knowledge of smart manufacturing solutions.
3. Identify new opportunities in smart industry.
4. Acquire skills to face future smart-industry challenges and opportunities.

Course Outcomes

1. Understand basic Industry 4.0 concepts.
2. Understand technologies and develop PLC programming knowledge.
3. Understand Industry 4.0 building blocks.
4. Explore robotics advances in the Industry 4.0 era.

CO-PO mapping as supplied

CO-PO mapping is reproduced from the supplied syllabus. The numerical mapping is retained as printed; blank cells are not silently converted into values.

Legend: 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Origin and concept	1
module-1	M1.02	Basic components	2
module-1	M1.03	Supportive technologies	4
module-1	M1.04	Proposed framework	2
module-1	M1.05	Smart-industry model sketch	2
module-2	M2.01	HMI, VR and AR	2
module-2	M2.02	Advanced robotics and 3-D printing	2
module-2	M2.03	Cloud computing and EAAS	2
module-2	M2.04	Connectivity solutions	2
module-2	M2.05	PLC and ladder logic	3
module-3	M3.01	Cyber-physical systems	2
module-3	M3.02	CPS characteristics	1
module-3	M3.03	Industrial Internet of Things	2
module-3	M3.04	Horizontal/vertical integration and RAMI 4.0	2
module-3	M3.05	IoT use cases and smart factories	2
module-3	M3.06	IoT and Industry 4.0 connectivity	2
module-4	M4.01	Recent robot components	2
module-4	M4.02	Advanced sensor technologies	2
module-4	M4.03	Internet of Robotic Things and cloud robotics	2
module-4	M4.04	Cognitive architecture for cyber-physical robotics	2

Module	Topic code	Topic	Hours
module-4	M4.05	Industrial robotic applications	2

Learning Roadmap

1. Origin, concepts and framework of Industry 4.0
CO1 · 11 hours

2. Technologies, connectivity and PLC programming
CO2 · 11 hours

3. Building blocks, CPS, IIoT and RAMI 4.0
CO3 · 11 hours

4. Robotics in Industry 4.0
CO4 · 10 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Origin, concepts and framework of Industry 4.0

Industry 4.0 connects cyber, physical, data and manufacturing systems. Its framework is built from smart machines, connectivity, data, automation and human decision support.

Hours: 11

CO1 · Bloom level: Understanding

Learning goals

- Explain origin and core idea.
- Identify components and supportive technologies.
- Prepare a smart-industry model sketch.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

– M1.01 — Origin and concept (1 hours)

Concept and explanation

Industry 4.0 describes the move toward connected, data-driven, flexible and intelligent production. It follows earlier industrial revolutions in mechanisation, mass production and automation.

Malayalam support: ഐ ഐ ഐ ഐ ഐ English technical terms, symbols, units, equations, and standard abbreviations ഐ ഐ ഐ ഐ.

Study points

- State the core idea.
- Relate Industry 4.0 to earlier stages.
- Give one manufacturing example.

– M1.02 — Basic components (2 hours)

Concept and explanation

Components include cyber-physical systems, sensors, actuators, networks, cloud services, analytics, robotics and human interfaces. Integration is more important than any single device.

Malayalam support: ഏകീകൃത സിസ്റ്റം ഘടകങ്ങൾ English technical terms, symbols, units, equations, and standard abbreviations ഉപയോഗിച്ച് എഴുതുക.

Study points

- List components.
- Explain their relationship.
- Identify data flow.

– M1.03 — Supportive technologies (4 hours)

Concept and explanation

Supportive technologies include IoT, cloud computing, AI/analytics, additive manufacturing, robotics, advanced sensors, mobile connectivity and digital twins or related digital models.

Malayalam support: ഓപ്പോർട്ട്യൂണിറ്റി ഘടകങ്ങൾ English technical terms, symbols, units, equations, and standard abbreviations ഉപയോഗിച്ച് എഴുതുക.

Study points

- Identify technologies.
- Match technology to use case.
- Explain enabling role.

– M1.04 — Proposed framework (2 hours)

Concept and explanation

A framework organises layers from physical assets and connectivity to data, applications, services and business decisions. It helps structure a smart-factory design.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉപയോഗിക്കുക.

Study points

- Explain framework layers.
- Trace a data-to-decision path.

– M1.05 — Smart-industry model sketch (2 hours)

Concept and explanation

A model sketch should show machines, sensors, PLC/control, network, data platform, analytics, operator interface and output or maintenance action.

Malayalam support: മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉപയോഗിക്കുക.

Study points

- Draw labelled blocks.
- Show direction of information flow.
- Connect a use case to the sketch.

Module summary: Industry 4.0 begins with a systems view: assets, data, connectivity, intelligence and people work together.

OFFICIAL MODULE 2

Technologies, connectivity and PLC programming

Smart manufacturing uses human-machine interfaces, robotics, additive manufacturing, cloud services and diverse communication technologies. PLC ladder logic links sensing to control.

Hours: 11

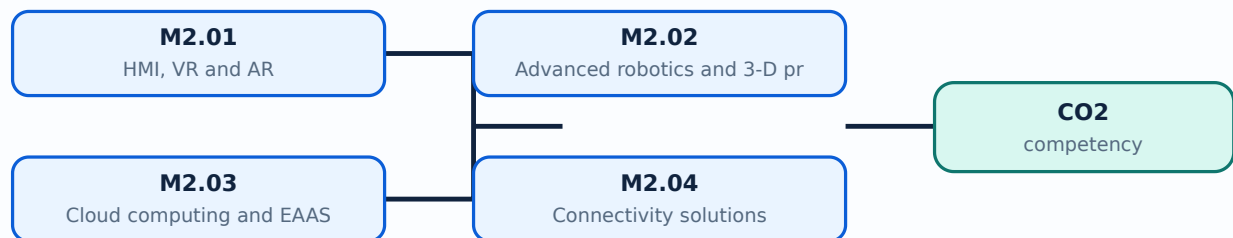
CO2 · Bloom level: Understanding

Learning goals

- Explain HMI, VR/AR, robotics and 3-D printing.
- Identify connectivity technologies.
- Outline PLC ladder programming.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

– M2.01 — HMI, VR and AR (2 hours)

Concept and explanation

Human-machine interfaces present status and controls. Virtual reality creates an immersive digital environment; augmented reality overlays information on the physical scene.

Malayalam support: ഏതെങ്കിലും മൂന്നു മലയാളം പദങ്ങൾക്ക് ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, ക്രമരേഖകൾ ഉൾപ്പെടെയുള്ള ഇംഗ്ലീഷ് പദങ്ങൾ ഉൾപ്പെടുത്തുക.

Study points

- Compare HMI, VR and AR.
- List industrial uses.
- Relate interface to operator decision.

– M2.02 — Advanced robotics and 3-D printing (2 hours)

Concept and explanation

Advanced robots perform autonomous or collaborative tasks; 3-D printing builds parts layer by layer. Both support flexible production and rapid prototyping.

Malayalam support: ഏതെങ്കിലും മൂന്നു മലയാളം പദങ്ങൾക്ക് ഇംഗ്ലീഷ് സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, ക്രമരേഖകൾ ഉൾപ്പെടെയുള്ള ഇംഗ്ലീഷ് പദങ്ങൾ ഉൾപ്പെടുത്തുക.

Study points

- State applications.
- Explain additive approach.
- Identify safety and quality needs.

Concept and explanation

Cloud computing provides scalable storage, processing and services. Equipment-as-a-Service supplies capability through a service model rather than only equipment ownership.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Explain cloud role.
- Define EAAS.
- Identify connectivity dependency.

Concept and explanation

Bluetooth, BLE, Bluetooth 5.0, ZigBee, ZigBee 3.0, Z-Wave, 6LoWPAN, RFID, Wi-Fi, mobile/cellular and SATCOM offer different range, power, rate and topology characteristics.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Compare technologies conceptually.
- Match technology to application.
- Mention security and interference.

Concept and explanation

A PLC reads inputs, executes a control program and updates outputs cyclically. Ladder logic uses contacts, coils, timers, counters and interlocks to represent industrial control.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Explain PLC scan cycle.
- Read a ladder rung.
- Design a simple start/stop interlock.

Engineering / workplace application: PLC control is used in conveyors, pumps, packaging and machine automation.

Module summary: Smart technologies become useful when reliable connectivity and deterministic control connect the physical process to digital services.

OFFICIAL MODULE 3

Building blocks, CPS, IIoT and RAMI 4.0

Cyber-physical systems combine computation, communication and physical processes. IIoT connects industrial assets, while integration frameworks describe information and process relationships.

Hours: 11

C03 · Bloom level: Understanding

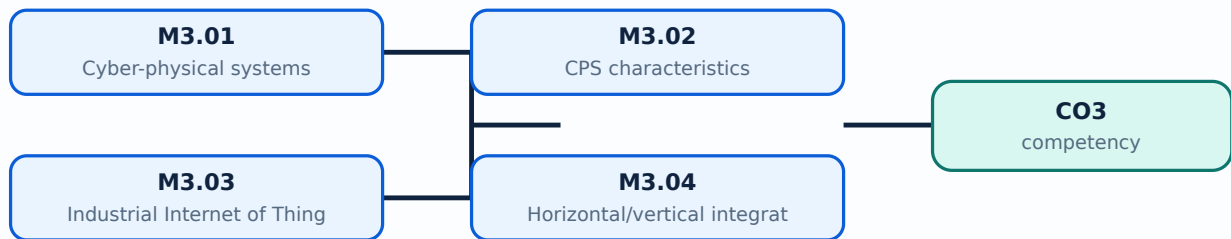
Learning goals

- Explain CPS and IIoT.
- Describe horizontal and vertical integration.

- Use RAMI 4.0 and smart-application concepts.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

– M3.01 — Cyber-physical systems (2 hours)

Concept and explanation

A CPS observes a physical process through sensors, computes decisions and acts through actuators while communicating with other systems. Feedback and timing are central.

Malayalam support: ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ ഞ English technical terms, symbols, units, equations, and standard abbreviations ഞ ഞ ഞ ഞ ഞ ഞ ഞ.

Study points

- Define CPS.
- Trace sensing-decision-actuation.
- Identify timing and safety needs.

Concept and explanation

Key characteristics include connectivity, computation, feedback, autonomy, distributed operation, adaptability, reliability and interaction with the physical environment.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- List characteristics.
- Relate characteristic to an industrial example.

Concept and explanation

IIoT connects industrial sensors, machines, controllers, networks, platforms and applications. It supports monitoring, optimisation, traceability and predictive maintenance.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Define IIoT.
- State use cases.
- Distinguish industrial from consumer context.

Concept and explanation

Horizontal integration links value-chain or production stages; vertical integration links field devices, control, operations and enterprise levels. RAMI 4.0 provides a reference architecture for organising assets, hierarchy and lifecycle.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Compare integration directions.
- Explain RAMI 4.0 purpose.
- Trace information across levels.

Concept and explanation

Smart factories, plants and applications use connected data for quality, maintenance, energy and production decisions. The use case should identify asset, data, action and benefit.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Construct a use-case chain.
- Identify measurable benefit.

M3.06 — IoT and Industry 4.0 connectivity (2 hours)

Concept and explanation

Connectivity enables CPS capabilities but introduces interoperability, cybersecurity, latency, reliability and data-governance requirements.

Malayalam support:   English technical terms, symbols, units, equations, and standard abbreviations  .

Study points

- List connectivity concerns.
- Relate network choice to control need.

Module summary: CPS and IIoT are the operating building blocks that connect industrial assets to integrated digital decisions.

OFFICIAL MODULE 4

Robotics in Industry 4.0

Robotics is becoming more connected, sensor-rich and data-driven. Cloud and cognitive capabilities extend robots beyond isolated repetitive motion.

Hours: 10

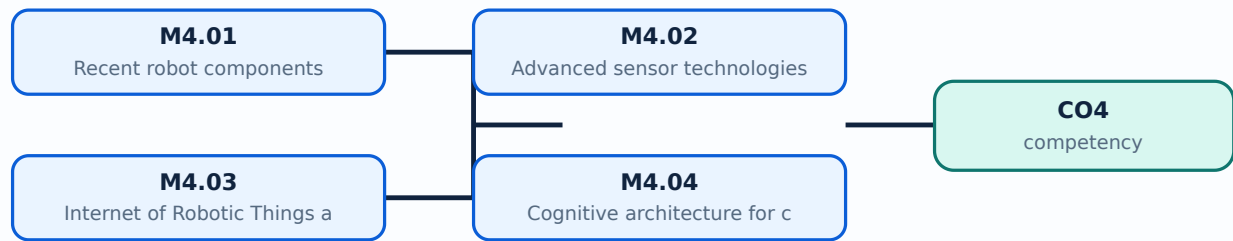
CO4 · Bloom level: Understanding

Learning goals

- Identify recent robot components.
- Explain advanced sensors and cloud robotics.
- Relate robotics to manufacturing, maintenance and assembly.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



– M4.01 — Recent robot components (2 hours)

Concept and explanation

Modern robots combine controllers, manipulators, drives, end effectors, safety systems, vision and communication interfaces. Components determine reach, payload, accuracy and collaboration capability.

Malayalam support: □ □□□□ □□□□□□□□□□ English technical terms, symbols, units, equations, and standard abbreviations □□□□□□□ □□□□□□□.

Study points

- Identify components.
- Relate component to function.
- State safety needs.

– M4.02 — Advanced sensor technologies (2 hours)

Concept and explanation

Force, torque, vision, proximity, lidar, inertial and other sensors allow robots to perceive position, objects, contact and environment.

Malayalam support: ഏതെങ്കിലും മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉൾപ്പെടെയുള്ളവ.

Study points

- Classify sensors.
- Match sensor to task.
- Explain sensor feedback.

– M4.03 — Internet of Robotic Things and cloud robotics (2 hours)

Concept and explanation

The Internet of Robotic Things connects robots to networks and other assets. Cloud robotics shares computation, learning, data and services, subject to latency and safety constraints.

Malayalam support: ഏതെങ്കിലും മലയാളം സാങ്കേതിക പദങ്ങൾ, ചിഹ്നങ്ങൾ, യൂണിറ്റുകൾ, സമവാക്യങ്ങൾ, and standard abbreviations ഉൾപ്പെടെയുള്ളവ.

Study points

- Define IoRT.
- Explain cloud-robot interaction.
- Identify latency concerns.

– M4.04 — Cognitive architecture for cyber-physical robotics (2 hours)

Concept and explanation

A cognitive architecture combines perception, world model, planning, decision, learning and action. It supports adaptive behaviour while retaining safety constraints.

Malayalam support: ഏകീകൃത മനോരൂപം English technical terms, symbols, units, equations, and standard abbreviations ഉപയോഗിക്കുക.

Study points

- List architecture functions.
- Trace perception to action.
- State the role of learning.

– M4.05 — Industrial robotic applications (2 hours)

Concept and explanation

Manufacturing, maintenance and assembly use robots for handling, welding, inspection, servicing and repetitive or hazardous operations. Selection considers payload, reach, accuracy, cycle time and safety.

Malayalam support: ഏകീകൃത മനോരൂപം English technical terms, symbols, units, equations, and standard abbreviations ഉപയോഗിക്കുക.

Study points

- List applications.
- Choose selection factors.
- Explain human-robot safety.

Module summary: Industry 4.0 robotics combines mechanical capability with sensing, connectivity, intelligence and safe industrial integration.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

[Open the module to view its labelled learning-map diagram.](#)

[Module 2: Technologies,](#)

[Open the module to view its labelled learning-map diagram.](#)

[Module 3: Building blocks,](#)

[Open the module to view its labelled learning-map diagram.](#)

[Module 4: Robotics in](#)

[Open the module to view its labelled learning-map diagram.](#)

Official Activities and Practical Requirements

Official activities / self-learning

- Prepare a labelled smart-industry model sketch.
- Write a ladder-logic sequence for a safe start/stop process.
- Select a connectivity technology for one industrial use case and justify it.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied syllabus is practical in scheme (L:0,T:0,P:3), but it lists Series Test entries. The exact institutional practical-evaluation mark distribution is not stated in the supplied PDF.

Module-wise Questions and Answers

module-1 — Origin, concepts and framework of Industry 4.0

1. Explain Origin and concept. [3 marks]

Industry 4.0 describes the move toward connected, data-driven, flexible and intelligent production. It follows earlier industrial revolutions in mechanisation, mass production and automation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Basic components. [3 marks]

Components include cyber-physical systems, sensors, actuators, networks, cloud services, analytics, robotics and human interfaces. Integration is more important than any single device.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Supportive technologies. [3 marks]

Supportive technologies include IoT, cloud computing, AI/analytics, additive manufacturing, robotics, advanced sensors, mobile connectivity and digital twins or related digital models.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Proposed framework. [3 marks]

A framework organises layers from physical assets and connectivity to data, applications, services and business decisions. It helps structure a smart-factory design.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Technologies, connectivity and PLC programming

5. Explain HMI, VR and AR. [3 marks]

Human-machine interfaces present status and controls. Virtual reality creates an immersive digital environment; augmented reality overlays information on the physical scene.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Advanced robotics and 3-D printing. [3 marks]

Advanced robots perform autonomous or collaborative tasks; 3-D printing builds parts layer by layer. Both support flexible production and rapid prototyping.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Cloud computing and EAAS. [3 marks]

Cloud computing provides scalable storage, processing and services. Equipment-as-a-Service supplies capability through a service model rather than only equipment ownership.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Connectivity solutions. [3 marks]

Bluetooth, BLE, Bluetooth 5.0, ZigBee, ZigBee 3.0, Z-Wave, 6LoWPAN, RFID, Wi-Fi, mobile/cellular and SATCOM offer different range, power, rate and topology characteristics.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Building blocks, CPS, IIoT and RAMI 4.0

9. Explain Cyber-physical systems. [3 marks]

A CPS observes a physical process through sensors, computes decisions and acts through actuators while communicating with other systems. Feedback and timing are central.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain CPS characteristics. [3 marks]

Key characteristics include connectivity, computation, feedback, autonomy, distributed operation, adaptability, reliability and interaction with the physical environment.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Industrial Internet of Things. [3 marks]

IIoT connects industrial sensors, machines, controllers, networks, platforms and applications. It supports monitoring, optimisation, traceability and predictive maintenance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Horizontal/vertical integration and RAMI 4.0. [3 marks]

Horizontal integration links value-chain or production stages; vertical integration links field devices, control, operations and enterprise levels. RAMI 4.0 provides a reference architecture for organising assets, hierarchy and lifecycle.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Robotics in Industry 4.0

13. Explain Recent robot components. [3 marks]

Modern robots combine controllers, manipulators, drives, end effectors, safety systems, vision and communication interfaces. Components determine reach, payload, accuracy and collaboration capability.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Advanced sensor technologies. [3 marks]

Force, torque, vision, proximity, lidar, inertial and other sensors allow robots to perceive position, objects, contact and environment.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Internet of Robotic Things and cloud robotics. [3 marks]

The Internet of Robotic Things connects robots to networks and other assets. Cloud robotics shares computation, learning, data and services, subject to latency and safety constraints.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Cognitive architecture for cyber-physical robotics. [3 marks]

A cognitive architecture combines perception, world model, planning, decision, learning and action. It supports adaptive behaviour while retaining safety constraints.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Origin and concept* with one relevant application. (5 marks)
2. **2.** Define or explain *Basic components* with one relevant application. (5 marks)
3. **3.** Define or explain *HMI, VR and AR* with one relevant application. (5 marks)
4. **4.** Define or explain *Advanced robotics and 3-D printing* with one relevant application. (5 marks)
5. **5.** Define or explain *Cyber-physical systems* with one relevant application. (5 marks)
6. **6.** Define or explain *CPS characteristics* with one relevant application. (5 marks)
7. **7.** Define or explain *Recent robot components* with one relevant application. (5 marks)
8. **8.** Define or explain *Advanced sensor technologies* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Origin, concepts and framework of Industry 4.0:** Industry 4.0 begins with a systems view: assets, data, connectivity, intelligence and people work together.
- **Technologies, connectivity and PLC programming:** Smart technologies become useful when reliable connectivity and deterministic control connect the physical process to digital services.
- **Building blocks, CPS, IIoT and RAMI 4.0:** CPS and IIoT are the operating building blocks that connect industrial assets to integrated digital decisions.
- **Robotics in Industry 4.0:** Industry 4.0 robotics combines mechanical capability with sensing, connectivity, intelligence and safe industrial integration.

Exam checklist

- Write the official terminology before adding explanation.

- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Origin and concept	Industry 4.0 describes the move toward connected, data-driven, flexible and intelligent production. It follows earlier industrial revolutions in mechanisation, mass production and automation.
Basic components	Components include cyber-physical systems, sensors, actuators, networks, cloud services, analytics, robotics and human interfaces. Integration is more important than any single device.
Supportive technologies	Supportive technologies include IoT, cloud computing, AI/analytics, additive manufacturing, robotics, advanced sensors, mobile connectivity and digital twins or related digital models.
Proposed framework	A framework organises layers from physical assets and connectivity to data, applications, services and business decisions. It helps structure a smart-factory design.
Smart-industry model sketch	A model sketch should show machines, sensors, PLC/control, network, data platform, analytics, operator interface and output or maintenance action.
HMI, VR and AR	Human-machine interfaces present status and controls. Virtual reality creates an immersive digital environment; augmented reality overlays information on the physical scene.
Advanced robotics and 3-D printing	Advanced robots perform autonomous or collaborative tasks; 3-D printing builds parts layer by layer. Both support flexible production and rapid prototyping.
Cloud computing and EAAS	Cloud computing provides scalable storage, processing and services. Equipment-as-a-Service supplies capability through a service model rather than only equipment ownership.
Connectivity solutions	Bluetooth, BLE, Bluetooth 5.0, ZigBee, ZigBee 3.0, Z-Wave, 6LoWPAN, RFID, Wi-Fi, mobile/cellular and SATCOM offer different range, power, rate and topology characteristics.

Term	Short meaning
PLC and ladder logic	A PLC reads inputs, executes a control program and updates outputs cyclically. Ladder logic uses contacts, coils, timers, counters and interlocks to represent industrial control.
Cyber-physical systems	A CPS observes a physical process through sensors, computes decisions and acts through actuators while communicating with other systems. Feedback and timing are central.
CPS characteristics	Key characteristics include connectivity, computation, feedback, autonomy, distributed operation, adaptability, reliability and interaction with the physical environment.
Industrial Internet of Things	IIoT connects industrial sensors, machines, controllers, networks, platforms and applications. It supports monitoring, optimisation, traceability and predictive maintenance.
Horizontal/vertical integration and RAMI 4.0	Horizontal integration links value-chain or production stages; vertical integration links field devices, control, operations and enterprise levels. RAMI 4.0 provides a reference architecture for organising assets, hierarchy and lifecycle.
IoT use cases and smart factories	Smart factories, plants and applications use connected data for quality, maintenance, energy and production decisions. The use case should identify asset, data, action and benefit.
IoT and Industry 4.0 connectivity	Connectivity enables CPS capabilities but introduces interoperability, cybersecurity, latency, reliability and data-governance requirements.
Recent robot components	Modern robots combine controllers, manipulators, drives, end effectors, safety systems, vision and communication interfaces. Components determine reach, payload, accuracy and collaboration capability.
Advanced sensor technologies	Force, torque, vision, proximity, lidar, inertial and other sensors allow robots to perceive position, objects, contact and environment.
Internet of Robotic Things and cloud robotics	The Internet of Robotic Things connects robots to networks and other assets. Cloud robotics shares computation, learning, data and services, subject to latency and safety constraints.
Cognitive architecture for cyber-physical robotics	A cognitive architecture combines perception, world model, planning, decision, learning and action. It supports adaptive behaviour while retaining safety constraints.
Industrial robotic applications	Manufacturing, maintenance and assembly use robots for handling, welding, inspection, servicing and repetitive or hazardous operations. Selection considers payload, reach, accuracy, cycle time and safety.

Official References and Resources

References listed in the supplied syllabus

1. Handbook of Industry 4.0 and SMART Systems, Diego Galar Pascual, Pasquale Daponte and Uday Kumar.
2. Industry 4.0: Managing the Digital Transformation, Alp Ustundag and Emre Cevikcan.
3. The Fourth Industrial Revolution, Klaus Schwab.
4. Industry 4.0: The Industrial Internet of Things, Alasdair Gilchrist.

Official learning resources listed

- <https://nptel.ac.in/courses/106/105/106105195/>
- <https://www.udemy.com/course/intro-to-industry-4/>

In-page Search

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